

Chapter 8

CELL BIOLOGY & BIOTECHNOLOGY



1. Fill in the blanks and complete the statements.

- a. Methods like artificial insemination and embryo transplant are mainly used for animal husbandry.
- b. Stem cell research is the revolutionary event in biotechnology after cloning.
- c. The disease related with the synthesis of insulin is Diabetes.
- d. Government of India has encouraged the people pisciculture for improving the productivity by launching NKM-16.

2. Match the pairs.

Interferon	Viral infection
Factor VIII	Haemophilia
Somatostatin	Dwarfness
Interleukin	Cancer
Insulin	Diabetes

3. Rewrite the following wrong statements after corrections.

- a. Changes in genes of the cells are brought about in non-genetic technique.

False. Changes in genes of the cells are brought about in genetic technique.

- b. Gene from *Bacillus thuringiensis* is introduced into soyabean.

False. Gene from *Bacillus thuringiensis* is introduced into cotton and brinjal.

4. Write short notes.

a. Biotechnology: Professional uses.

- i. Use of various abilities of microbes like yoghurt production from milk and alcohol from molasses.
- ii. Use of productivity of cells e.g., Production of antibiotics and vaccines with the help of specific cells.
- iii. Use of bio-molecules like DNA and proteins in human welfare.
- iv. Development of plants, animals and products of desired quality by gene manipulation. e.g., Production of growth hormone with the help of genetically modified bacteria.
- v. Use of genetic and non-genetic technique.
- vi. Non-genetic biotechnology involves use of either cell or tissue. e.g., tissue culture, production of hybrid seeds.



b. Importance of medicinal plants.

- i. India has been gifted with a great biodiversity. Indian citizens have a strong and humble relation with nature.

- ii. We have a great tradition of ayurveda that cures the diseases with the help of natural sources.
- iii. Medicinal plants find applications in food industry, pharmaceuticals, agriculture and cosmetics.
- iv. There are infinite medicinal plants which come together and form a branch of medical science called ayurveda.



5. Answer the following questions in your own words.

a. Which products produced through biotechnology do you use in your daily life?

- i. One of the simplest uses of biotechnology that we practice at home is making curd and buttermilk.
- ii. The primary type of biotechnology is used in the fermentation process when making food items such as bread, idli-dosa, dhokla, etc.
- iii. Today, various types of cheese, paneer, yogurt, energy drinks, and more are produced with the assistance of biotechnology.
- iv. We consume these products in our daily lives.

- v. Seedless grapes, papaya, and watermelons are readily available in the market nowadays.
- vi. Violet cabbage, yellow capsicum, and exotic vegetables used in salads are also products of biotechnology.
- vii. Vaccines, antibiotics, and injections of human insulin are regularly used in many households.

b. Which precautions will you take during spraying of pesticides?

- i. Pesticides are, in fact, a type of poison. This poison enters the food web through water and food, and bio-magnification occurs.
- ii. Various pesticides such as DDT, malathion, and chlorpyrifos are dangerous.
- iii. Precautions to be taken include:
 - a. Apply only the recommended dose.
 - b. Cover your nose with a cloth before spraying.
 - c. Always wear gloves while handling the equipment.
 - d. Avoid spillage of pesticides.
 - e. Do not inhale the pesticide.
 - f. Read the instructions carefully.
 - g. Do not apply against the direction of the wind, during rainy or sunny days.
 - h. Containers and buckets used for spraying should not be used for domestic purposes.
 - i. Prevent the entry of animals and workers into the field during spraying.

c. Why some of the organs in human body are most valuable?

- i. Various organs in human body become less efficient or completely functionless, due to various reasons like ageing, accidents, infections etc.
- ii. Life of such persons become difficult, miserable, uncomfortable. Sometimes death may also occur.
- iii. Many organs in human body remain functional for certain period even after death occurs under specific conditions.
- iv. These organs can be transplanted on donation.
- v. Hence, certain organs like heart, kidneys, eyes, liver, heart valves, skin, etc. are more valuable.



d. Explain the importance of fruit processing in human life?

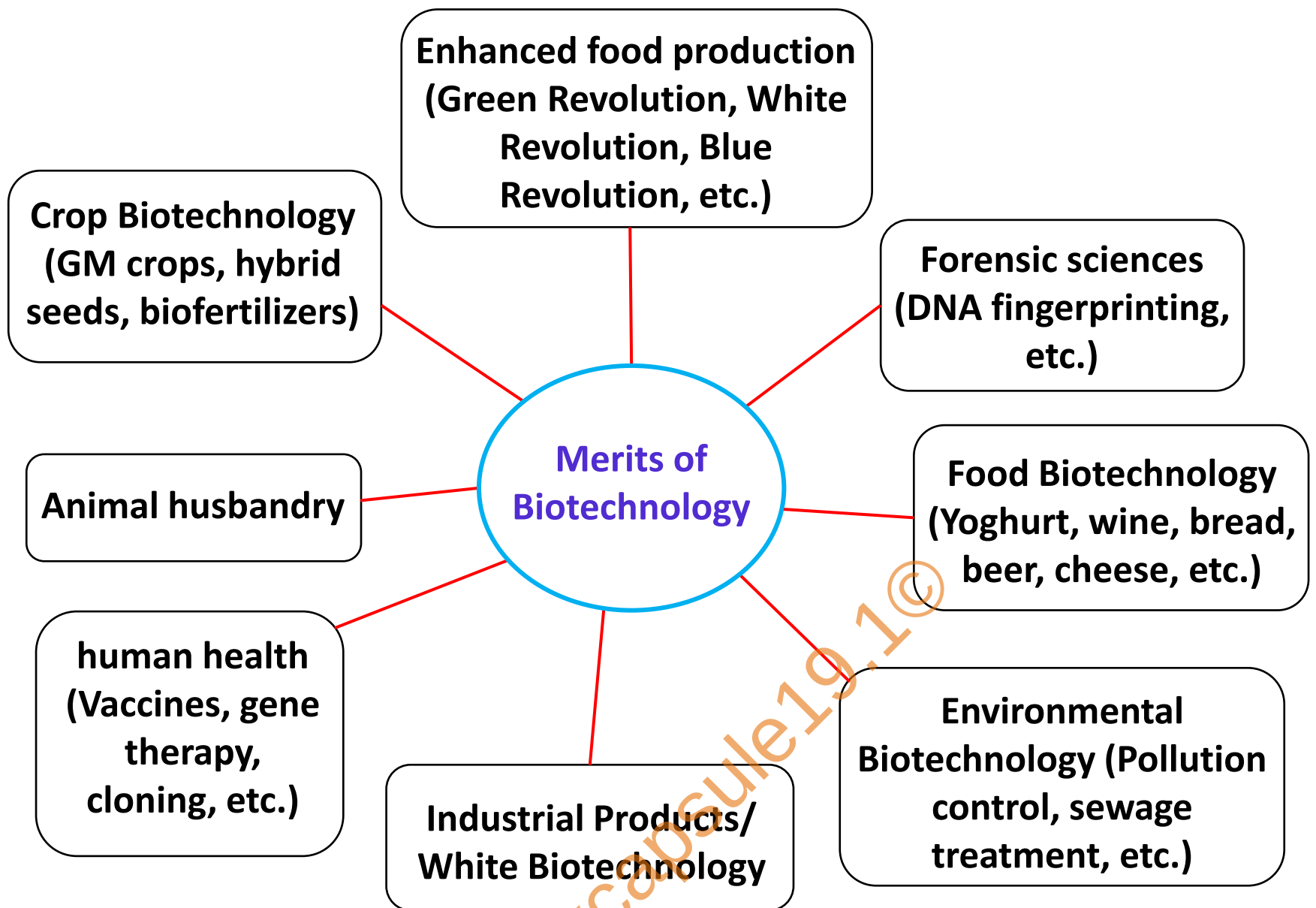
- i. Fruits are perishable agricultural produce.
- ii. Fruits contain vitamins, proteins, minerals, and dietary fibers.
- iii. There is a need to process fruits in order to significantly increase their shelf life.
- iv. Various products are prepared from fruits, such as chocolates, jams, juices, and jellies, which are used daily.
- v. Fruit processing is done in such a way that it can be used throughout the year.

e. Explain the meaning of vaccination.

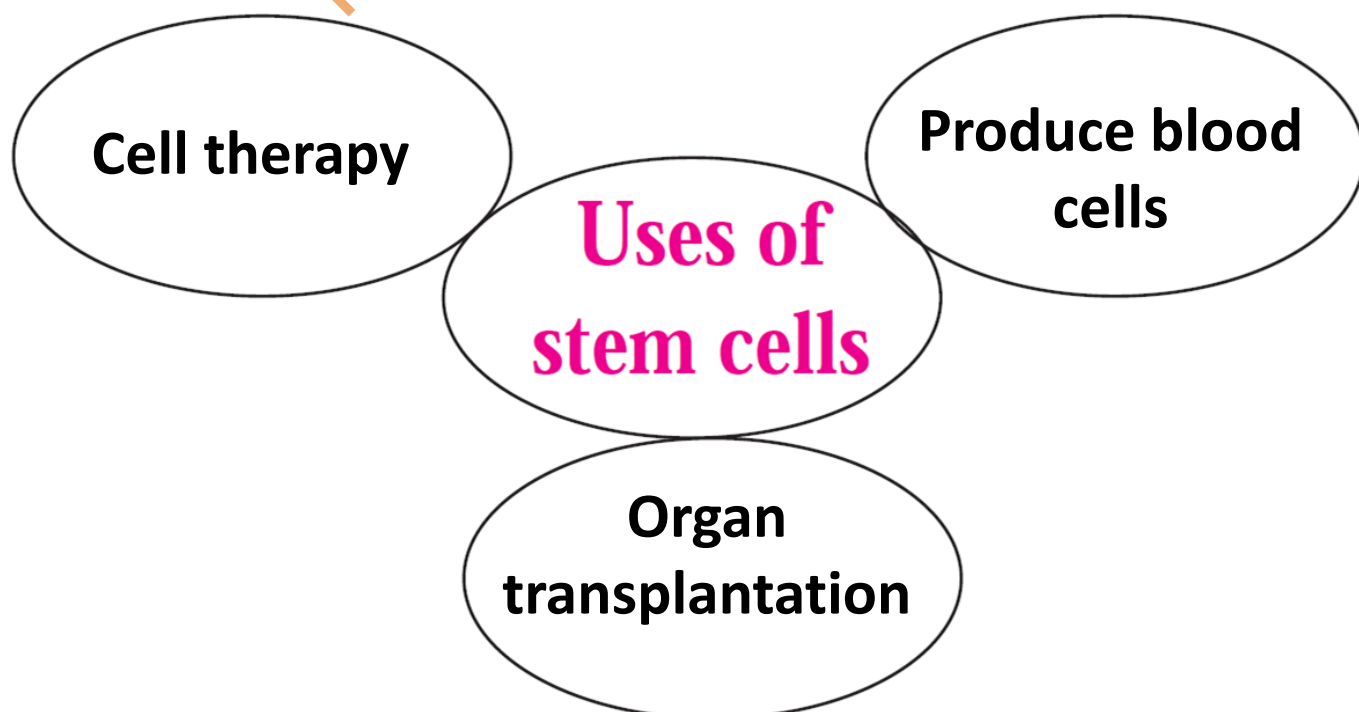
- i. Vaccine is the 'antigen' containing material given to acquire either permanent or temporary immunity against a specific pathogen or disease.
- ii. Traditionally, vaccines were prepared with the help of pathogens which may be completely or partially killed.
- iii. However due to this, there were chances of contracting the disease in case of certain persons.
- iv. Hence, as an alternative, scientists tried to artificially produce vaccines with the help of biotechnology.
- v. For this purpose, scientists produced the antigen in laboratory with the help of gene isolated from the pathogen and used it as vaccine. Thus, safer vaccines are being produced.
- vi. Now, proteins which act as antigen are injected in pure form instead of injecting the killed or semi-killed pathogens.
- vii. These proteins keep the persons away from the disease by keeping the immune system active.
- viii. Vaccines produced with the help of biotechnology are more thermostable and remain active for longer duration. E.g., vaccines of polio, hepatitis etc.



6. Complete the following chart.



7. Write the correct answer in blank circles.



8. Identify and complete the following correlations.

a. **Insulin : Diabetes :: Interleukin : Cancer.**

Insulin is used to treat diabetes, whereas Interleukin is used to treat cancer.

b. **Interferon : Viral infection :: Erythropoietin : Anaemia.**

Erythropoietin is used for treating conditions like anaemia, whereas interferons are used to treat viral infections.

c. **Somatostatin : Dwarfness :: Factor VIII : Haemophilia.**

Factor VIII is used for treating condition like haemophilia, whereas somatostatin is used for treating dwarfism.

d. **White revolution : Dairy :: Blue revolution : Aquatic organisms.**

White revolution involves improved production of dairy and dairy products whereas, Blue revolution involves improved production of Aquatic organisms.

9. Write a comparative note on usefulness and harmfulness of biotechnology.

Usefulness of biotechnology:

- i. Biotechnology has revolutionized medicine, leading to life-saving drugs and treatments.
- ii. It enhances agriculture by increasing crop yields and reducing the need for pesticides.
- iii. Biotech solutions tackle environmental issues like pollution cleanup and renewable biofuels.

- iv. Industries benefit from biotechnology with the production of enzymes, bio-based materials, and bioplastics.
- v. Biotech tools advance scientific research in genetics, genomics, and microbiology.

Harmful effects of Biotechnology are as follows.

- i. Biotechnology can harm the environment through factors like GMOs and soil degradation.
- ii. Health risks can emerge from the development of antibiotic-resistant bacteria and concerns about genetically modified foods.
- iii. Ethical dilemmas arise in areas like human cloning, gene editing, and designer babies.
- iv. Biotechnology can worsen economic inequality, limiting access to advanced treatments and technologies.
- v. Biodiversity can suffer due to the reduced genetic diversity of biotech crops and the displacement of traditional agriculture.



INTEXT QUESTIONS

1. What is cell?

Cell is defined as a basic structural, and functional and unit of life.

2. What is tissue? Which are the functions of tissue?

A group of cells having the same origin, same structure and same function is called as tissue. There are five basic functions of tissues: (a) secretion (b) movement (c) strength (d) excretion (e) communication.

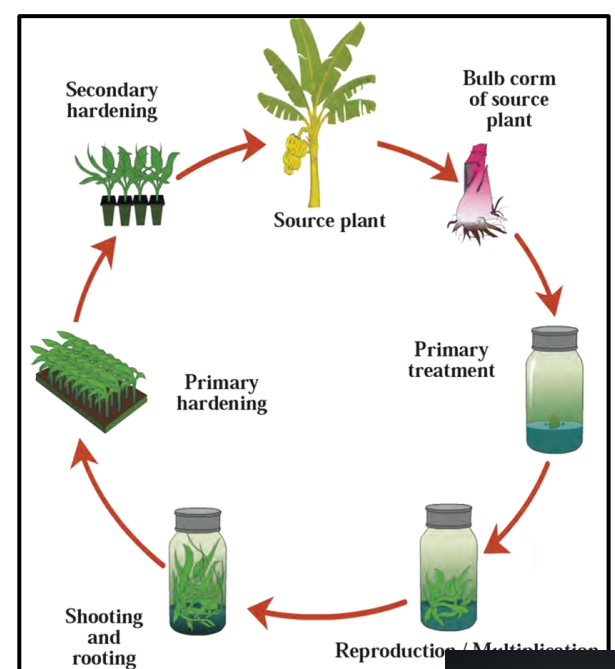
3. Which technique in relation to tissues have you studied in earlier classes?

Tissue culture is the technique studied in earlier classes.

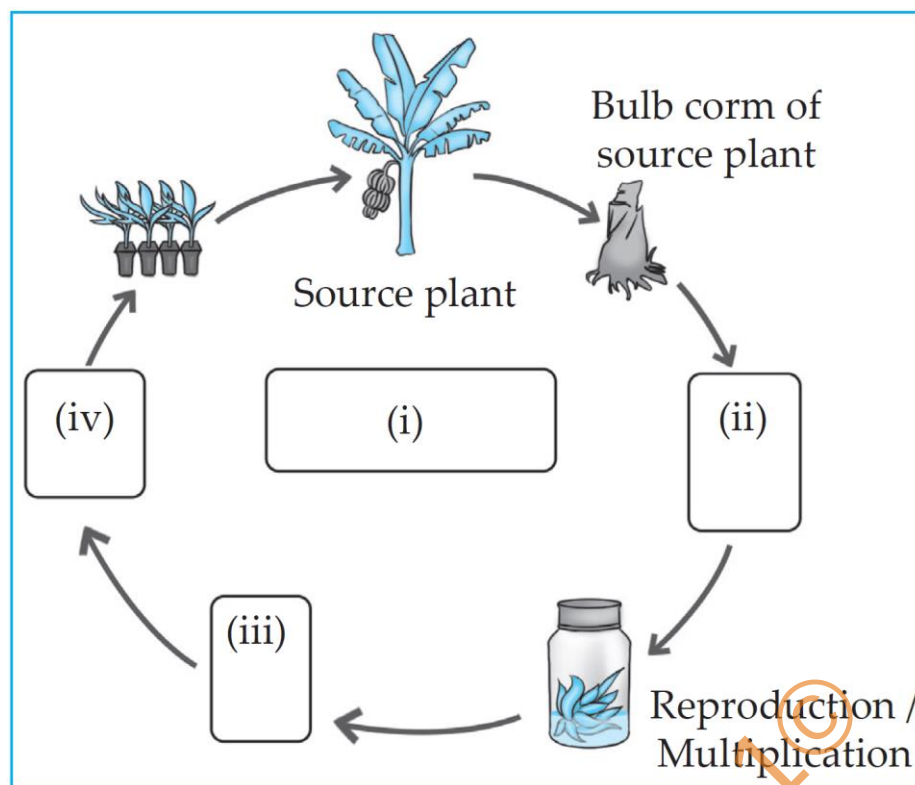
4. Which are the various processes in tissue culture?

The various processes used in tissue culture are as follows:

- Selection of source plant.
- Removal of Bulb corm of source plant.
- Primary treatment of the part.
- Reproduction and Multiplication.
- Shooting and Rooting
- Primary hardening.
- Secondary hardening.



5. Assign names in the figure given below. Explain the various stages those are kept blank.

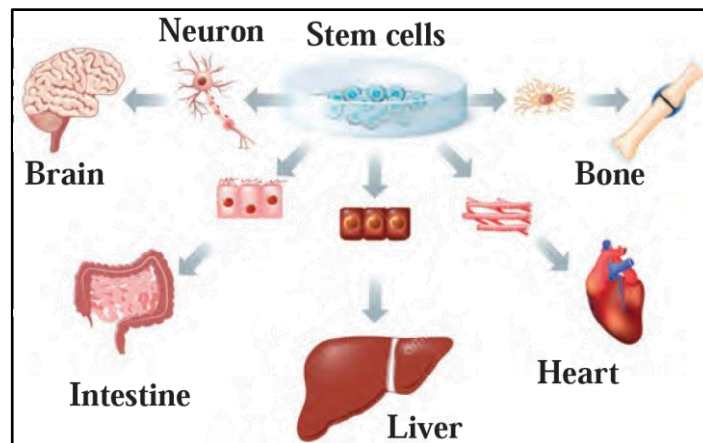


- (i) Various processes in tissue culture
- (ii) Primary treatment: It is the initiation phase. Plant tissue is sterilized In vitro and initiated into culture.
- (iii) Shooting and rooting: It is the phase of root formation. Hormones are introduced to induce rooting and for the formation of a complete plantlet.
- (iv) Primary hardening: Hardening is gradually exposing the plants to adjust to the environment conditions.

6. Just like the grafting in plants, is the organ transplantation possible in humans?

Yes, human organ transplantation is possible. It is usually the primary treatment for end-stage organ failure, such as in the case of the liver and heart.

7. Discuss about stem cells and organ transplantation in the class with the help of figures given below.



- i. In the given diagram, we can observe that stem cells can form a variety of new cell types that can combine to form tissues and organs.
- ii. With advances in biotechnology, if provided with specific chemicals/stimulus and optimum conditions, stem cells may develop into neurons, brain, intestines, liver, heart and bone.
- iii. These valuable organs could then be used for organ transplantation to help people in need, i.e., in whom the organs have become functionless due to certain conditions.

8. What is biotechnology?

The technique of bringing about improvements in living organisms by artificial genetic changes and by hybridization for the welfare of human beings are together called biotechnology.



9. In which various fields, the biotechnology has been useful?

Biotechnology has been useful in medicine, agriculture, environmental science, industry, scientific research, food production, energy production, environmental conservation, biomedical engineering, and forensics.

10. What is the impact of biotechnology on agriculture and other related fields?

- i. Genetically modified crops are being produced by introducing changes in DNA of natural crops.
- ii. It has become possible to increase the per hectare yield irrespective of the limitations of crop-land area.
- iii. Expenses on disease control have minimized since development of resistant varieties.
- iv. Due to development of fast fruit setting varieties, yield per annum has been increased.
- v. Development of stress resistant varieties which can withstand variable temperature, water stress, changing fertility of soil, etc. has become possible.
- vi. Biofertilizers, bioinsecticides have been developed with the help of biotechnology which prevent the soil from being polluted and increase the yield.
- vii. With the help of biotechnology, plants and microorganisms are used for absorption or destruction of toxic chemicals and harmful pollutants. This is called bioremediation.

11. Give five examples of each of the fruiting and flowering plants developed through tissue culture and mention their benefits.

Fruits and flowers developed through tissue culture are:

Fruits: Pineapple, Banana, Pomegranate, Papaya Apple, etc.

Flowers: Marigold, Rose, Shevanti, Lily, Gerbera daisy, etc.

Benefits of tissue cultured fruits and flowers:

- i. These are almost pest and disease resistant.
- ii. Mature plant is obtained in less time.
- iii. More plants with desired characteristics like bigger size, colourful fruits and disease resistance, etc. are obtained.
- iv. Planting is possible all through the year.
- v. Rare and endangered varieties also can be grown by tissue culture.



12. Which new species of the rice have been developed in India?

- i. BPT 2270 (Bhavapuri Sannalu)
- ii. BPT 2231 (Akshaya)
- iii. Pusa 1460
- iv. RPB10226

13. Which fruit processing industries you observe in your surrounding ?

What is their effect ?

Fruit Processing Industries:

- i. Juice Production
- ii. Canned Fruits
- iii. Jam and Jelly Production
- iv. Frozen Fruits
- v. Dried Fruit Processing

Effects:

Positive Effects:

- i. Processed fruit products ensure fruits are available regardless of the season.
- ii. Surplus or imperfect fruits are used in processing, reducing food waste.
- iii. Ready-to-eat fruit options are convenient for consumers.
- iv. Fruit processing industries provide jobs and economic opportunities in the food sector.

Negative Effects:

- i. High-temperature processing can lead to nutrient loss in some products.
- ii. Some processed fruit products may contain added sugars or preservatives, raising health concerns.
- iii. Packaging and waste management in these industries have environmental implications.



Previous Year Questions

1. Read the following extract and answer the questions that follow:

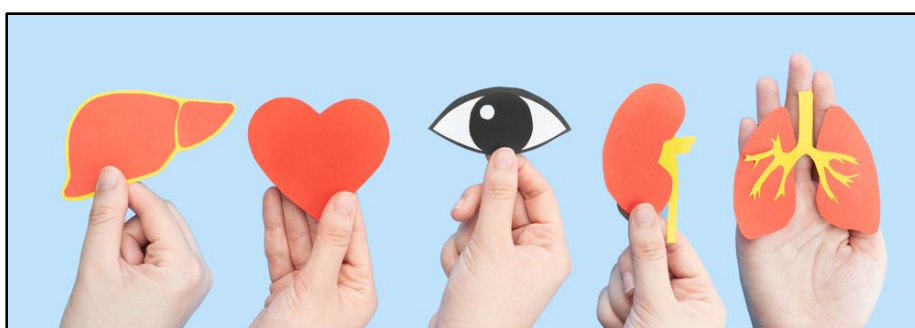
A Liberal view behind the concept of organ and body donation is that after death our body should be useful to other needful persons so that their miserable life would become comfortable. Awareness about these concepts is increasing in our country and people are voluntarily donating their bodies. Life of many people can be saved by organ and body donation. Blinds can regain their vision. Life of many people can be rendered comfortable by donation of organs like liver, kidneys, heart, heart valves, skin etc. Similarly, body can be made available for research in medical studies. Many government and social organizations are working towards increasing the awareness about body donation.

a. What is the liberal view behind the organ and body donation?

A liberal view behind the concept of organ and body donation is that after death our body should be useful to other needful persons so that their miserable life would become comfortable.

b. Name any four organs that can be donated.

Organs like liver, kidneys, heart, heart valves, skin etc.



2. What is biotechnology?

The technique of bringing about improvements in living organisms by artificial genetic changes and by hybridization for the welfare of human beings are together called biotechnology.

Explain any two commercial applications of it.

- i. Crop Biotechnology: Biotechnology is used in agricultural field to improve yield and variety, i.e., hybrid seeds, genetically modified crops, herbicide tolerant plants and biofertilizers.
- ii. Animal Husbandry: Artificial insemination and embryo transfer are used to improve quality and quantity of animal products.

3. Complete the following table:

Sunflower

Deinococcus radiodurans

Absorb the arsenic from the soil

Useful for cleaning the hydrocarbon and oil pollutants from soil and water

4. What is Biotechnology?

The technique of bringing about improvements in living organisms by artificial genetic changes and by hybridization for the welfare of human beings are together called biotechnology.

Give one use of Biotechnology.

Use of various abilities of microbes like yoghurt production from milk and alcohol from molasses

Give one commercial use of Biotechnology.

Crop Biotechnology: Biotechnology is used in agricultural field to improve yield and variety, i.e., hybrid seeds, genetically modified crops, herbicide tolerant plants and biofertilizers.

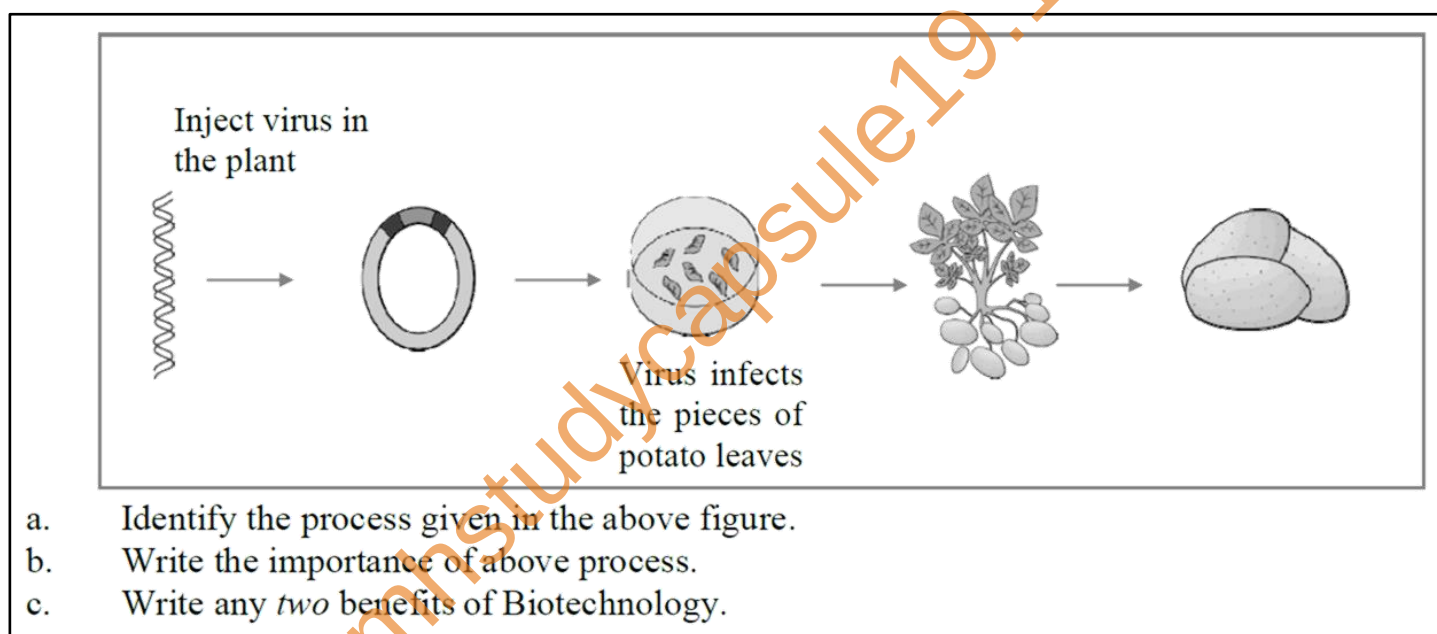
Write two bacterial examples of biofertilizer.

Rhizobium, Azotobacter

Write two names of crops genetically developed.

BT cotton, BT brinjal

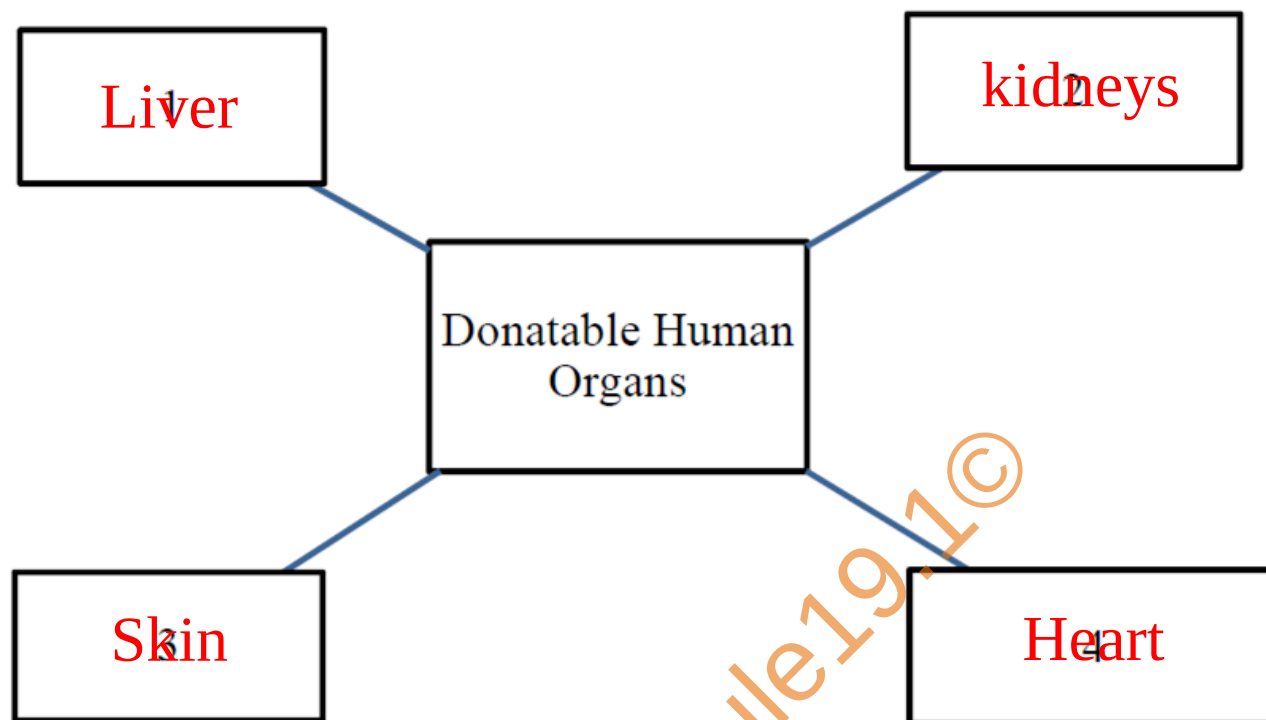
5. Observe the figure and answer the following questions:



- Production of Transgenic potatoes
- Work on production of edible vaccines is in progress and presently, potatoes are being produced with the help of biotechnology. These potatoes are called as transgenic potatoes. These potatoes will act against bacteria like *Vibrio cholerae*, *Escherichia coli*. Consumption of these raw potatoes generates the immunity against cholera and the disease caused due to *E. coli*.

- c. It has become possible to increase the per hectare yield irrespective of the limitations of crop-land area. Expenses on disease control have minimized since development of resistant varieties.

6. Complete the chart:



7. Which precautions will you take during spraying of pesticides? –

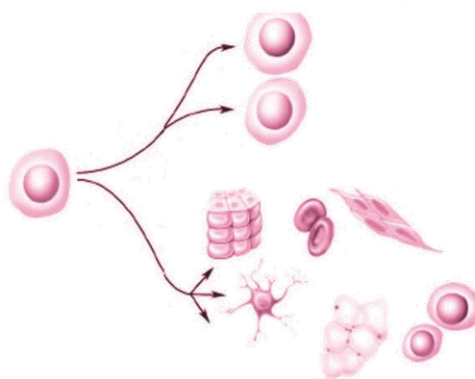
Exercise Question 5 - b

8. Define genetically modified crops and give any two examples.

Crops developed with desired characters by integrating foreign gene within their genome are called as genetically modified crops.

1. BT cotton
2. BT brinjal

9. Which process is shown in the given diagram?



Process shown in the diagram is Stem cells therapy.

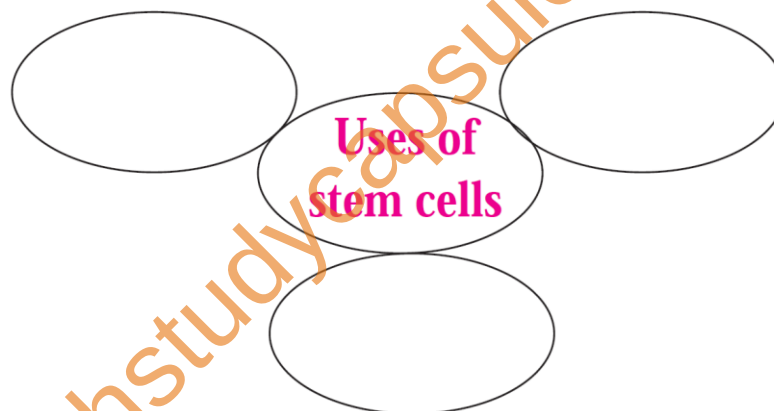
What is the importance of this process?

Stem cells are used to replace the dead cells in case of conditions like diabetes, myocardial infarction, Alzheimer's disease, Parkinson's disease, etc. To produce blood cells required in conditions like anaemia, thalassemia, leukemia, etc.

Which organs can be transplanted by this process?

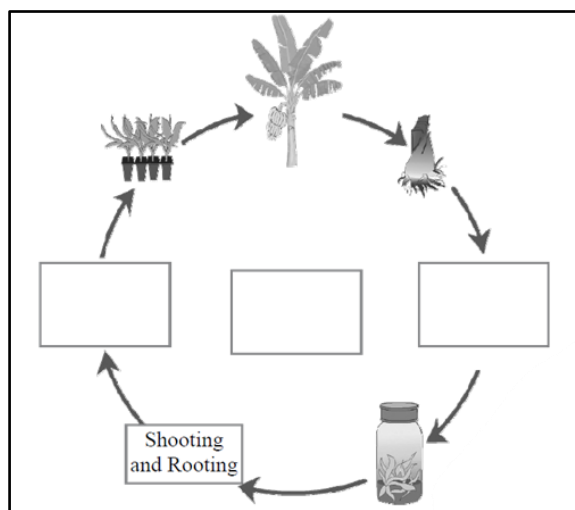
In case of failure of organs like kidney and liver, those can be produced with the help of stem cells and transplanted.

10. Write the correct answer in blank circles. – Exercise Question 7

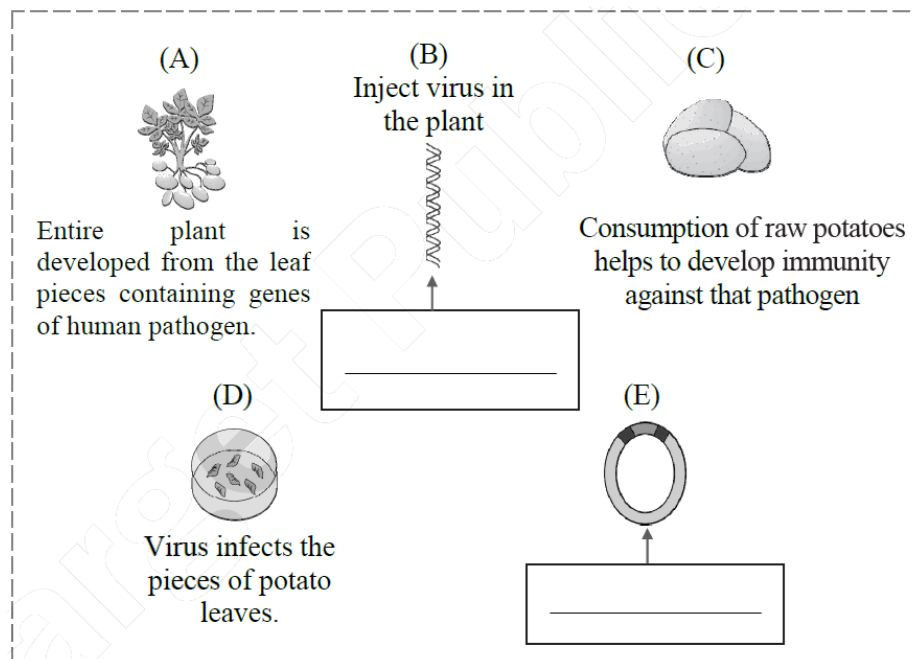


11. Complete the diagram showing various processes in tissue culture.

– Intext Question



12. Rearrange and write proper alphabetical sequence only and complete the given blanks:



(B) Isolation of desired gene from human pathogen

(E) Transgenic plant virus

(D)

(A)

(C)